

PRACTICUM-I PROJECT PLAN

ThePetsVault

AI-Powered Pet Care & Marketplace Ecosystem

Project Vision: One trusted digital ecosystem connecting pet buying/selling, pet sitting, veterinary care and health records, and pet food/accessories around a unified pet profile - enhanced by practical AI decision-support and trust features.

Program	Master of Science in Software Project Management (MS-SPM)
Course	Practicum-I
Institution	National University of Computer & Emerging Sciences (FAST-NUCES), Karachi Campus
Semester	Fall 2026
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Document Control

Field	Detail
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Status	For review and sign-off

Revision History

Version	Date	Description	Author
1.0	Sep 2026	Initial project plan baseline for supervisor sign-off	Student Team

Approval / Supervisor Sign-Off

By signing below, the supervisor confirms that the proposed project direction, scope baseline and practicum planning approach are suitable to proceed. Approval does not constitute acceptance of final implementation results; material scope changes remain subject to formal review.

Decision	Supervisor Name	Signature	Date	Comments
☐ Approved ☐ Approved with Conditions ☐ Rework Required	Muhammad Kamran			

EXECUTIVE SUMMARY

ThePetsVault is proposed as a unified digital ecosystem for Pakistan's pet-care market. Its core proposition is not simply to create another pet classifieds application; it is to connect four services that are currently fragmented across informal channels and separate platforms: a verified pet buy-and-sell marketplace, pet sitting/boarding, veterinary consultation and appointment management with longitudinal pet health records, and an e-commerce marketplace for pet food and accessories. Every service is connected to a single pet profile and a common trust layer comprising identity verification, ratings/reviews, secure platform-mediated payments, moderation, and auditable transaction records.

The project is planned as a 12-month academic master project. Practicum-I establishes the validated business case, requirements baseline, architecture, UX prototype, core platform foundations, and proof-of-concept AI components. Practicum-II completes the integrated product, testing, pilot launch and closure. This staged plan reconciles the earlier 6-9 month MVP estimate with the additional time required for academic validation, controlled pilot operation, documentation and post-launch stabilization.

AI is integrated where it improves trust, discovery and service efficiency without replacing licensed veterinary judgment. The proposed AI scope includes personalized matching and recommendation, listing/fraud risk scoring, a grounded support and veterinary triage assistant, and smart product/sitter recommendations. Advanced diagnostic prediction and autonomous medical advice are explicitly excluded from the MVP. The AI layer will begin with transparent rule-based and content-based methods and can evolve toward machine-learning models as sufficient data becomes available.

The plan uses a hybrid Agile-PMI governance model: Agile delivery and Kanban/Scrum practices for iterative software development, while scope, budget, risk, stakeholder approvals, quality gates and change control are managed through formal SPM/PMI artifacts. The planning baseline uses a PKR 4.80 million academic budget estimate, a 12-month schedule, measurable success criteria, a structured WBS, risk register, communication matrix and Earned Value Management framework.

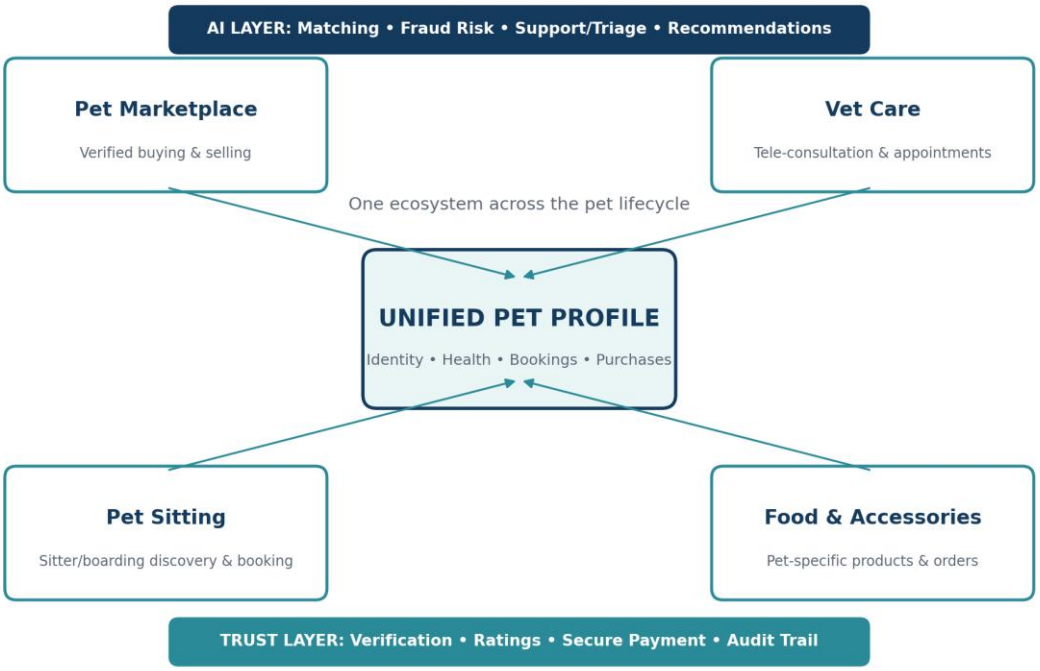


Figure 1. ThePetsVault ecosystem: four integrated service pillars connected by a unified pet profile, AI layer and trust layer.

Table of Contents

Section	Title
1	Business Statement
2	Feasibility Analysis
3	Critical Success Factors
4	Project Goal, Purpose, Aim & Objectives
5	Project Deliverables
6	Project Scope
7	Assumptions, Constraints & Dependencies
8	Stakeholder Management
9	Expert & Domain Validation Plan
10	Work Breakdown Structure
11	Master Project Schedule
12	Risk Management
13	Human Resource Management
14	Time-Phased Budget
15	Earned Value Management Framework
16	Communication Management
Appendix A	AI Integration & Governance
Appendix B	MoSCoW Product Backlog
Appendix C	Definition of Done & Quality Gates
Appendix D	Glossary & Source Materials

1. BUSINESS STATEMENT

Business context, opportunity and strategic drivers

1.1 Organization Background

ThePetsVault is conceptualized as a Pakistan-focused technology venture designed to formalize and simplify the companion-animal ecosystem. The platform's vision is to become a trusted digital destination for pet ownership by combining commerce, care and services rather than forcing users to manage each need through disconnected apps, social-media groups, individual clinics and informal referrals.

The baseline product is organized around four core service pillars: (1) verified pet buying and selling, (2) pet sitting/boarding and short-term care, (3) veterinary consultation, appointment booking and pet medical/vaccination history, and (4) pet food and accessories. A unified pet profile ties these pillars together, allowing pet-related history, preferences, bookings and approved health data to move with the owner across the ecosystem.

> **Positioning statement:** ThePetsVault is a trusted pet lifecycle marketplace - not only a classifieds site, not only a vet app, and not only an online pet store.

1.2 Organization Structure & Explanation

A balanced matrix structure is recommended because the project requires simultaneous software engineering, project governance and domain expertise. The core practicum team remains accountable for delivery, while specialist roles such as veterinary review, payment/KYC integration, security testing and business onboarding are engaged at defined gates. This is appropriate for an academic startup-style project because it provides expertise without assuming a large permanent organization.

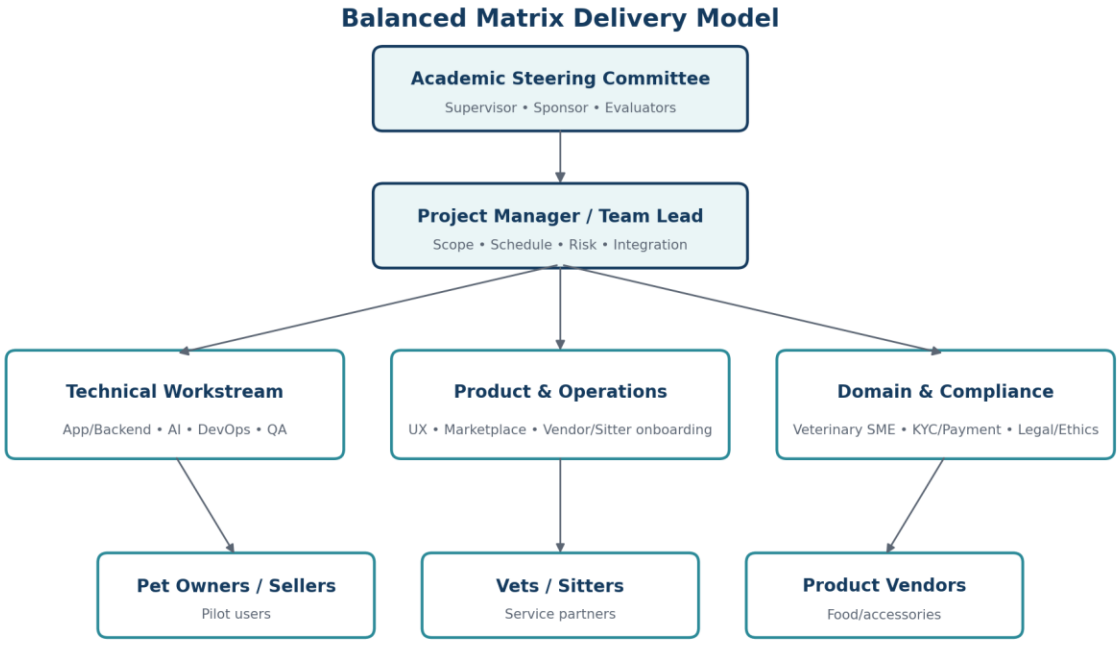


Figure 2. Proposed balanced matrix project organization.

1.3 Problem Statement

The principal problem is fragmentation and low trust. A pet owner may use informal social-media listings to buy a pet, a separate clinic or WhatsApp contact for veterinary care, unrelated e-commerce stores for supplies, and friends or unverified contacts for pet sitting. These disconnected experiences create duplicated effort, limited continuity of health information, inconsistent service quality and a greater risk of fraud or poor animal-care outcomes.

- Pet transactions often rely on informal communication and direct transfers, creating limited payment protection and weak seller accountability.
- Pet medical and vaccination records are usually dispersed across paper cards, chat history and clinic systems rather than attached to a portable pet profile.
- Owners who travel have difficulty finding verified, rated and available sitters or boarding hosts.
- Food and accessory purchases are rarely personalized to the actual pet profile or linked to veterinarian-approved recommendations.
- Existing solutions generally address one vertical, leaving the user to coordinate the full lifecycle manually.

1.4 Opportunity

The opportunity is to shift from a passive classifieds model toward an integrated service marketplace with an explicit trust architecture. The platform can create recurring engagement: a user who acquires or registers a pet can later book veterinary care, store vaccination history, purchase supplies and arrange sitting without leaving the ecosystem. This cross-vertical retention is the core strategic advantage.

1.5 Business Drivers

Driver	Why it Matters
Digital trust	Demand for verified identities, transparent profiles, ratings and secure in-app transaction flows.
Convenience & consolidation	One account and pet profile replacing several disconnected channels.
Growth of urban pet care	Concentration of digitally active pet owners, vets, sitters and delivery networks in Karachi, Lahore and Islamabad.
Recurring revenue	Marketplace commissions, sitting booking fees, vet consultation fee share, e-commerce margin and optional promoted listings.
Data continuity	A longitudinal pet profile improves experience and enables responsible personalization.
AI-enabled differentiation	Matching, moderation and guided support reduce user effort while improving trust and platform efficiency.

1.6 Target Market

The initial geographic focus is urban Pakistan, with Karachi recommended for the controlled pilot because it provides the concentration required for a two-sided marketplace. Primary demand-side users include first-time/repeat pet buyers, existing pet owners seeking routine or remote veterinary support, frequent travellers requiring care for pets, and regular purchasers of pet food/accessories. Supply-side stakeholders include verified sellers/breeders, independent pet sitters/boarding hosts, licensed veterinarians/clinics and pet-product vendors.

2. FEASIBILITY ANALYSIS

Operational, technical, economic and schedule feasibility

2.1 Operational Feasibility - Rating: 9/10

Operational feasibility is high because the solution consolidates user journeys that already occur manually: pet discovery, veterinary care, travel-time pet care and recurring supply purchases. The main operational challenge is supply-side quality; credible sellers, vets, sitters and product vendors must be onboarded before demand is scaled.

2.2 Technical Feasibility - Rating: 8/10

Technical feasibility is supported by established technologies and provider integrations instead of custom payment, video or identity infrastructure. Flutter supports Android/iOS; role-based web portals serve partners and administrators; a Node.js/NestJS-style REST backend and PostgreSQL manage transactional workflows; and a Python/FastAPI service isolates AI logic. Cloud hosting, notifications, maps and a commercial video SDK further reduce implementation risk.

Feasibility safeguard: Production access to CNIC/KYC and payment services is an external dependency. The practicum will use authorized provider sandboxes or a simulated adapter if live approval is not available. The project will not build a custom payment gateway or claim direct NADRA access without formal authorization.

2.3 Economic Feasibility - Rating: 8/10

Economic feasibility is supported by multiple revenue streams: pet-sale and sitting commissions, veterinary consultation fee share, e-commerce margins/listing fees and promoted visibility. The academic planning baseline is PKR 4.80 million, within the estimated investment range in the proposal. Commercial payback is treated as a later-stage objective dependent on marketplace liquidity and repeat usage.

2.4 Schedule Feasibility - Rating: 8/10

A single-semester production build is high risk because the solution spans four verticals plus trust and AI. A 12-month master plan makes the schedule feasible: Practicum-I covers requirements, architecture, core prototype and AI proof-of-concept; Practicum-II completes integration, testing, pilot and deployment, with a controlled MVP targeted around Month 9.

Feasibility Type	Rating	Primary Success Driver
Operational	9/10	Strong pain-point fit and clear multi-role workflows
Technical	8/10	Modular architecture and proven third-party SDK/API usage
Economic	8/10	Multiple revenue streams and controlled MVP budget
Schedule	8/10	12-month phased plan with narrow Practicum-I baseline

3. CRITICAL SUCCESS FACTORS

CSF 1 - Trustworthy onboarding

Seller, sitter, vet and vendor profiles must be verifiable and auditable; suspicious accounts require manual review.

CSF 2 - Marketplace liquidity

A credible minimum supply of pets, sitters, veterinarians and products must exist before public demand campaigns.

CSF 3 - Safe pet-care workflows

Pet sitting and veterinary flows require clear responsibilities, emergency contacts, cancellation rules and escalation paths.

CSF 4 - Reliable pet profile & health records

Vaccination and medical history must be accurate, permission-controlled and easy for authorized vets/owners to update.

CSF 5 - Responsible AI

AI must improve discovery and triage without making autonomous diagnoses or unsafe health claims. Results must be explainable and overrideable.

CSF 6 - Payment and integration reliability

Transaction status, refunds/cancellations, video sessions, notifications and identity checks must fail safely when third-party services are unavailable.

CSF 7 - User experience

Search, listing, booking, consultation and checkout flows must be simple enough for non-technical pet owners and service providers.

CSF 8 - Security & privacy

CNIC/KYC references, health data, location and payment-related information require strict access controls, encryption and audit logs.

CSF 9 - Quality discipline

No critical/high-severity defects at pilot; security and end-to-end flows must pass UAT.

CSF 10 - Controlled scope

Advanced medical AI, proprietary payment rails, owned boarding facilities, and nationwide rollout remain outside the MVP unless formally approved.

4. PROJECT GOAL, PURPOSE, AIM & OBJECTIVES

4.1 Project Goal

To deliver a validated and scalable digital platform that makes pet ownership safer and easier by bringing pet commerce, care, health records and services into one trusted ecosystem.

4.2 Project Purpose

The purpose is to reduce fragmentation and trust gaps in the Pakistani pet ecosystem while demonstrating a commercially viable, technically feasible and academically defensible marketplace model for an MS Software Project Management practicum.

4.3 Project Aim

To design, prototype, develop and pilot ThePetsVault with four integrated service pillars, a unified pet profile, role-based partner portals, secure platform-mediated transactions and practical AI decision-support.

4.4 Project Objectives

Objective Area	Target
Functional	Implement pet buying/selling listings, sitter discovery/booking, vet appointment/tele-consultation, pet health/vaccination records, product commerce, ratings/reviews, notifications and admin moderation.
Trust & Safety	Provide verified onboarding workflow, listing moderation, dispute/cancellation handling, audit logs and secure payment integration through an approved provider.
AI	Deliver an AI recommendation/matching engine, grounded support/triage assistant and listing risk-scoring prototype with measurable evaluation criteria.
Technical	Use modular APIs, relational data models, RBAC, secure cloud deployment, CI/CD and provider adapters so external services can be replaced without redesigning the core system.
Business	Validate a sustainable multi-stream revenue model and a Karachi-first supply onboarding strategy.
UX	Reduce time and effort required to find trusted pets, services, vets and relevant products while preserving clear consent and transparency.
Academic	Produce complete SPM artifacts: scope baseline, WBS, schedule, risk plan, budget, EVM framework, communication plan, quality evidence and supervisor sign-offs.

4.5 Practicum-I Exit Criteria

Practicum-I exits when the scope/requirements baseline and user journeys are signed; architecture, data model and interactive UX prototype are approved; the core authentication, pet-profile and marketplace prototype demonstrates payment/KYC adapters in sandbox or mock mode; and the AI proof-of-concept plus updated risk, schedule and budget baselines are approved for Practicum-II.

5. PROJECT DELIVERABLES

ID	Deliverable	Description
D1	Mobile Pet Owner Marketplace App	Flutter app supporting user registration, pet profiles, pet listings/search, bookings, consultation access, orders and notifications.
D2	Partner & Admin Web Portals	Responsive web interfaces for veterinarians, product vendors, sitters/service partners and administrators.
D3	Unified Pet Profile & Health Record Module	Owner-controlled pet identity, vaccination history, allergies/notes, consultation history, appointments and uploaded documents with role-based access.
D4	Pet Buy/Sell Marketplace Module	Verified listings, search/filter, favorites, in-app communication, health/verification badges, reporting and transaction workflow.
D5	Pet Sitting / Boarding Module	Sitter profiles, service area, availability calendar, booking request, pricing, owner instructions, check-in updates, ratings and emergency-contact capture.
D6	Veterinary Care Module	Vet discovery, appointment calendar, online consultation session, consultation notes, prescriptions/uploads where appropriate, and follow-up scheduling.
D7	Pet Food & Accessories Module	Product catalog, vendor inventory, cart, checkout, order tracking and pet-profile-based recommendation hooks.
D8	Trust, Payment & Moderation Layer	Role verification, secure provider-integrated payment flow, cancellation/refund state machine, audit trail, reporting and moderation queues.
D9	AI Recommendation & Matching Engine	Content-based/rule-based recommendations for pets, sitters and products, designed to evolve to ML after sufficient usage data.
D10	AI Support & Vet-Triage Assistant	Grounded conversational assistant using approved platform/veterinary knowledge; classifies urgency and routes users to a vet without providing a diagnosis.
D11	AI Listing Risk / Scam Detection	Risk score using account/listing signals, suspicious pricing, duplicate/perceptual-image checks and behavior rules; high-risk content is escalated to human moderation.
D12	Cloud, Security & DevOps Baseline	Environment configuration, CI/CD, logs, backups, encryption, secrets handling, monitoring and VAPT/security testing evidence.
D13	QA/UAT & Pilot Evidence	Test plan, traceability matrix, defect log, security results, UAT sign-offs and Karachi pilot report.
D14	Project Management & Academic Documentation	Project charter, BRD/SRS/FSD, WBS, Gantt, risk register, budget, EVM reports, communication plan, change log, user manuals and closure report.

5.1 AI Deliverable Acceptance Metrics

AI Capability	MVP Technique	Target Measure

AI Capability	MVP Technique	Target Measure
Recommendation/matching	Rules + content-based similarity; user preference feedback	Precision@5 or evaluator relevance ≥ 70% on curated test set
Vet triage/support	RAG over vet-reviewed knowledge + red-flag rules	≥ 90% correct routing on urgent/non-urgent benchmark; zero autonomous diagnosis claims
Listing risk score	Rules/anomaly features + duplicate image hash	Recall ≥ 80% on seeded suspicious cases; false-positive rate ≤ 15%
AI service performance	FastAPI service with caching/fallback	Median response ≤ 2.5 seconds in test environment; graceful fallback if AI unavailable

6. PROJECT SCOPE

6.1 Scope Overview

The scope baseline prioritizes a working, integrated MVP rather than maximum feature breadth. The project must demonstrate the end-to-end lifecycle: verified user onboarding, pet profile creation, pet marketplace interaction, sitter booking, veterinary appointment/online consultation, product order, platform-mediated payment status and AI-assisted discovery/support.

6.2 In-Scope Functional Scope

- User registration, login, OTP/email verification and role-based profiles.
- Pet profiles with species/breed/age/sex/weight, vaccination records, medical notes, documents and owner permissions.
- Pet listing creation, search/filter, favorites, moderation, report listing and buyer/seller messaging.
- Pet sitting/boarding service listing, availability, booking, owner instructions, check-in updates, review/rating and cancellation flow.
- Veterinarian profile, appointment scheduling, tele-consultation via integrated video SDK, consultation notes and follow-up booking.
- Food/accessory product catalog, vendor portal, cart/checkout, order status and ratings.
- Payment gateway integration through approved/sandbox provider; commission calculation and transaction state tracking.
- KYC/identity verification adapter using authorized provider or simulated practicum service.
- Admin dashboard for users, listings, bookings, consultations, orders, disputes, risk flags and analytics.
- AI recommendation, grounded assistant/triage and listing risk scoring as defined in Section 5.1.
- Notifications through push/email/SMS adapter.
- Karachi pilot with controlled participants and documented feedback.

6.3 Out-of-Scope / Future Work

- Autonomous veterinary diagnosis, prescriptions generated by AI, or replacement of a licensed veterinarian.
- Predictive disease diagnosis from pet photos or medical images.
- Building a proprietary payment gateway or directly holding regulated customer funds without a licensed partner.
- Guaranteed direct NADRA integration; only authorized provider access may be used.
- Company-owned pet boarding facilities, warehouses or delivery fleet.
- International expansion during the practicum baseline.
- Trading of protected wildlife, illegal species or any transaction prohibited by applicable law.
- Full insurance underwriting or claims platform.
- Advanced computer-vision breed recognition and predictive health analytics (candidate future enhancements).
- NGO rescue/adoption marketplace as a core module; may be considered after MVP if separately approved.

6.4 Success Criteria

Area	Success Measure
Scope	100% of approved Must-have scope delivered; no unapproved feature added to baseline.
Schedule	Major milestones within ±10% schedule variance; Practicum-I exit criteria achieved by Month 6.
Cost	Cost variance within ±10% of approved academic BAC unless change request is approved.
Quality	≥95% planned test pass rate; zero open critical/high defects at pilot go/no-go.

Area	Success Measure
Performance	Core API p95 ≤ 2.0 seconds for standard operations under defined test load; AI median response ≤2.5 seconds.
Security	No Critical/High VAPT findings unresolved at pilot; RBAC/access-control tests passed.
AI	Targets in Section 5.1 met and AI output clearly labelled as decision-support.
User Acceptance	≥80% UAT scenario pass rate in first round and 100% critical scenarios passed after fixes.
Pilot	Controlled Karachi pilot demonstrates at least one successful end-to-end flow in each of the four service pillars.
Academic	Supervisor/panel sign-off on plan, prototype evidence, final report and project closure artifacts.

6.5 Scope Control

MoSCoW prioritization and formal change control will protect the practicum baseline. Any new request is evaluated for business value, academic value, effort, risk and effect on the critical path. A change that affects approved scope, cost by more than 5%, schedule by more than one week, data/privacy obligations or AI medical boundaries requires supervisor/project sponsor approval before implementation.

7. ASSUMPTIONS, CONSTRAINTS & DEPENDENCIES

7.1 Assumptions

- The core team can access representative pet owners, a veterinary SME and potential service partners for requirements validation and UAT.
- External integrations can provide sandbox/test access or be represented by approved adapters/mocks for academic demonstration.
- Owners will provide explicit consent before vets or sitters access pet-specific information; clinical AI content will be reviewed by a veterinary SME.
- Karachi can supply a controlled pilot cohort, and the project will not custody regulated funds or unnecessary national-identity data.

7.2 Constraints

Constraint	Implication
Academic calendar & team size	Practicum dates and a lean student team require strict prioritization, cross-training and schedule buffers.
External approvals	Live KYC/payment/video-provider production approval may delay integration; sandbox/mock adapters are required as fallback.
Data availability & AI	Limited behavioral data requires transparent rule/content-based methods before any ML evolution.
Veterinary safety & privacy	AI cannot diagnose; health, identity and location data require consent, minimum-data design and escalation to licensed care.
Budget & market liquidity	The PKR 4.80M academic baseline constrains paid services, while sufficient credible supply must exist before broad acquisition.

7.3 Major Dependencies

Dependency	Required For
Payment gateway / aggregator	Sandbox/live onboarding, callback/webhook reliability and refund/cancellation support.
KYC provider	Authorized identity-verification service or approved practicum mock adapter.
Video consultation SDK	Agora/Twilio or equivalent reliability and pricing.
Maps / geolocation	Location discovery for nearby vets/sitters and service-area validation.
Firebase/SMS/email	Authentication and booking/consultation/order notifications.
Veterinary SME	Health-record design, triage red-flags, knowledge-base approval and UAT.
Supply partners	Enough sellers, sitters, vets and product vendors for pilot usefulness.
Cloud hosting	Stable environment, storage, backups, monitoring and deployment pipelines.

8. STAKEHOLDER MANAGEMENT

8.1 Stakeholder Register

Stakeholder	Power	Interest	Primary Concern
Practicum Supervisor / Evaluators	High	High	Scope approval, academic quality, milestone sign-off
Project Team	High	High	Delivery, artifacts, technical implementation
Pet Owners / Buyers	Medium	High	Usability, trust, records, bookings, purchases
Pet Sellers / Breeders	Medium	High	Listings, verification, transactions, reputation
Pet Sitters / Boarding Hosts	Medium	High	Availability, booking, service standards, safety
Veterinarians / Clinics	High	High	Clinical workflow, health records, triage safety, consultation
Product Vendors	Medium	Medium	Catalog/inventory, orders, commission
Technology / Payment / QA Partners	High	Medium	Integration availability, certification, security and quality gates

8.2 RACI Matrix

Activity	Supervisor	PM/Lead	Dev/AI	Vet SME	QA/Sec	Partners
Scope & requirements	A	R	R	C	C	I
Architecture	I	A	R	C	C	C
AI safety design	I	A	R	A/C	C	I
Core development	I	A	R	C	C	I
KYC/payment integration	I	A	R	I	C	C
QA / VAPT / UAT	A	R	C	R	R	C
Pilot go/no-go	A	R	C	C	C	I
Project closure	A	R	C	C	C	I

R = Responsible, A = Accountable, C = Consulted, I = Informed.

9. EXPERT & DOMAIN VALIDATION PLAN

The source template includes a named industry expert section; no expert identity was supplied for ThePetsVault. The plan therefore defines the validation role without inventing a person. The final submission should replace the placeholders below once the supervisor approves an appropriate expert.

Validation Role	Recommended Profile	What Will Be Validated
Veterinary Domain Expert	Licensed veterinarian / clinic owner	Health record fields, red-flag triage rules, consultation workflow, AI knowledge-base safety
Software/Architecture Expert	Senior software engineer / solution architect	Architecture, API boundaries, scalability, security and implementation feasibility
Project Management Expert	PMP/SPM practitioner	WBS, schedule realism, risk register, budget and communication artifacts
Payment/KYC Advisor	Fintech/payment integration professional	Payment/KYC workflow feasibility and regulatory boundaries

9.1 Expert Sign-off Record

Expert	Area	Name / Organization	Date	Signature / Evidence
Expert 1	Veterinary			
Expert 2	Architecture / SPM			

10. WORK BREAKDOWN STRUCTURE

The WBS decomposes the project into manageable work packages aligned with the 12-month master schedule. The detailed WBS below is a management baseline; sprint-level user stories will be managed in Jira/Kanban without changing the approved WBS unless scope is formally changed.

WBS	Work Package / Task	Key Output	Owner	Timing
1.0	Project Initiation	Charter, team, kickoff, stakeholder register	PM	M1
1.1	Validate project vision	Confirm 4-pillar value proposition and AI scope	PM/Team	M1
1.2	Stakeholder identification	Supervisor, users, vets, sitters, sellers, vendors, providers	PM	M1
1.3	Project charter & approval	Objectives, authority, constraints and sign-off	PM	M1
2.0	Research & Business Validation	Market/competitor review, personas, business model	PM/Business	M1-M2
2.1	Persona interviews	Pet owners, travelling owners, buyers/sellers	Business	M1
2.2	Supply interviews	Vets, sitters/boarding, vendors	Business	M1-M2
2.3	Business model validation	Revenue, cost, pilot assumptions	PM	M2
3.0	Requirements & Scope Baseline	BRD/SRS, process flows, acceptance criteria	PM/BA	M1-M2
3.1	Marketplace requirements	Listing, search, moderation, transaction states	BA/Dev	M1-M2
3.2	Pet sitting requirements	Availability, booking, instructions, reviews, safety	BA	M2
3.3	Vet/health requirements	Appointments, video, records, permissions	BA/Vet	M2
3.4	Commerce requirements	Catalog, inventory, cart, order, vendor	BA	M2
3.5	AI requirements	Matching, triage guardrails, risk scoring, metrics	AI/Vet	M2
3.6	Non-functional requirements	Security, privacy, performance, availability	Architect/QA	M2
4.0	UX & Solution Architecture	Wireframes, architecture, data model, API specs	UX/Architect	M2-M3
4.1	Information architecture	Navigation and role journeys	UX	M2
4.2	Interactive prototype	Owner, seller, sitter, vet, vendor, admin	UX	M2-M3
4.3	Data model	Users, pets, health, listings, bookings, products, transactions	Backend	M3
4.4	API contract	REST endpoints, auth, events/webhooks	Backend	M3
4.5	AI architecture	RAG, feature store, model/service boundaries	AI	M3
5.0	Platform Foundations	Repos, CI/CD, auth, RBAC, base services	Dev/DevOps	M3-M4
5.1	Cloud environments	Dev/test/UAT configuration and monitoring	DevOps	M3
5.2	Authentication & RBAC	Owner/seller/sitter/vet/vendor/admin permissions	Backend	M3-M4
5.3	Unified pet profile	Pet master data, documents and consent	Backend/App	M4

WBS	Work Package / Task	Key Output	Owner	Timing
5.4	Notifications	Push/email/SMS adapter	Backend	M4
6.0	Pet Marketplace Development	Listings, search, moderation and messaging	Dev	M4-M6
6.1	Listing create/edit	Pet details, photos, health/verification badges	App/Backend	M4-M5
6.2	Search & filters	Species, breed, city, price, verified status	App/Backend	M5
6.3	Messaging & reporting	In-app communication and abuse/report flows	Dev	M5-M6
6.4	Transaction state flow	Offer/confirm/payment/cancel/complete	Backend	M5-M6
7.0	Trust, KYC & Payment	Provider adapters, risk controls, audit trail	Backend/QA	M4-M6
7.1	KYC adapter	Sandbox/mock identity verification abstraction	Backend	M4-M5
7.2	Payment adapter	JazzCash/Easypaisa/aggregator sandbox integration	Backend	M5-M6
7.3	Dispute/cancellation states	Refund and escalation workflow	Backend/PM	M6
7.4	Audit logs	Security-sensitive event logging	Backend	M5-M6
8.0	Pet Sitting / Boarding	Provider profile, availability and booking	Dev	M5-M7
8.1	Sitter profile & verification	Services, location, price, experience	App/Backend	M5
8.2	Availability calendar	Dates, capacity, blackout periods	Backend/App	M6
8.3	Booking & check-in	Request, accept, instructions, updates	Dev	M6-M7
8.4	Ratings & safety	Reviews, emergency contacts, incident reporting	Dev/PM	M7
9.0	Veterinary Care & Health Records	Appointments, video, health history	Dev/Vet	M5-M7
9.1	Vet discovery/profile	Specialization, clinic, availability	Dev	M5-M6
9.2	Appointment booking	Slot management and reminders	Dev	M6
9.3	Video consultation	SDK integration and session lifecycle	Dev	M6-M7
9.4	Health/vaccine records	Vet/owner permissions and updates	Backend/Vet	M6-M7
10.0	Food & Accessories Commerce	Catalog, vendor, cart and order	Dev	M6-M8
10.1	Vendor/catalog	Products, variants, prices, inventory	Web/Backend	M6-M7
10.2	Cart/checkout	Pet-linked cart and payment	App/Backend	M7
10.3	Order tracking	Status, notifications, review	Dev	M7-M8
11.0	AI & Decision Support	Matching, assistant and risk scoring	AI/Dev	M5-M8
11.1	Recommendation baseline	Content/rule matching for pets/sitters/products	AI	M5-M6
11.2	RAG knowledge base	Approved FAQs/pet-care guidance ingestion	AI/Vet	M6
11.3	Vet triage guardrails	Red-flag rules, routing, disclaimers	AI/Vet	M6-M7

WBS	Work Package / Task	Key Output	Owner	Timing
11.4	Listing risk scoring	Price anomalies, account signals, duplicate images	AI/Backend	M6-M7
11.5	Evaluation harness	Offline metrics, benchmark dataset, logs	AI/QA	M7-M8
12.0	Integration & QA	SIT, regression, security, performance	QA/Dev	M7-M9
12.1	Functional test suite	Traceability to acceptance criteria	QA	M7-M8
12.2	Integration testing	Payments, KYC, video, notifications, AI	QA/Dev	M8
12.3	Performance testing	API, DB, AI latency and load	QA/DevOps	M8-M9
12.4	Security/VAPT	Access control, OWASP, secrets, data exposure	QA/Security	M8-M9
13.0	UAT & Pilot Readiness	Training, UAT rounds, go/no-go	PM/QA	M8-M9
13.1	UAT scenarios	End-to-end flows across four pillars	PM/QA	M8
13.2	UAT execution	Users, vet SME, partners, supervisor demo	PM	M8-M9
13.3	Pilot data/privacy review	Consent, retention, incident process	PM/Security	M9
14.0	Karachi Pilot	Controlled pilot and feedback	PM/Team	M9
14.1	Supply onboarding	Pilot sellers, sitters, vets, vendors	Business	M8-M9
14.2	Pilot launch	Limited users and monitored transactions	PM	M9
14.3	Pilot metrics	Usage, errors, disputes, AI quality, feedback	PM/AI	M9
15.0	Hardening & Launch Preparation	Feedback fixes, documentation, readiness	Team	M10-M11
15.1	Prioritized fixes	Resolve pilot defects and usability issues	Dev/QA	M10
15.2	Operations runbooks	Admin, support, incident, data restore	PM/DevOps	M10
15.3	Partner training	Vet/vendor/sitter/admin guides	PM	M10-M11
15.4	Deployment approval	Go-live checklist and sign-off	Supervisor/PM	M11
16.0	Deployment & Support	Release, monitoring, controlled support	DevOps/PM	M11
16.1	Production deployment	Cloud release and smoke tests	DevOps	M11
16.2	Post-launch monitoring	Availability, incidents, AI logs, feedback	Team	M11
17.0	Closure & Defense	Handover, final report and lessons learned	PM/Team	M12
17.1	Project closure report	Scope/cost/schedule/quality summary	PM	M12
17.2	Academic defense package	Demo, presentation, evidence repository	Team	M12
17.3	Lessons learned/archive	Retrospective, repository and handover	PM	M12

11. MASTER PROJECT SCHEDULE

The schedule below is an indicative baseline beginning in Fall 2026. Exact academic dates will be adjusted to the supervisor's practicum calendar. The master plan preserves the proposal's 6-9 month MVP expectation by targeting a controlled pilot in Month 9, while using Months 10-12 for hardening, deployment, academic evidence and closure.

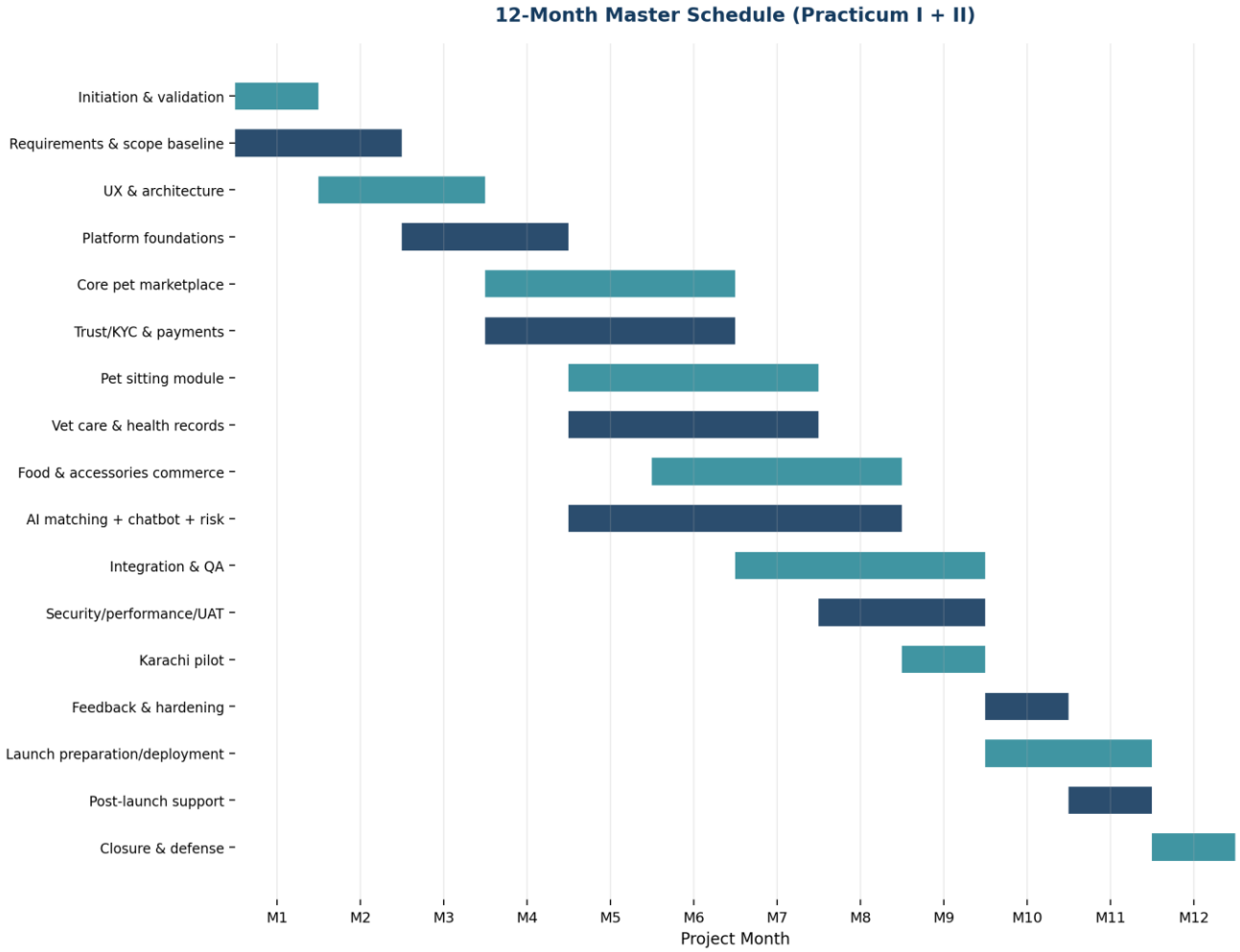


Figure 3. High-level 12-month master Gantt baseline.

11.1 Major Milestones

Milestone	Deliverable / Gate
M1	Project charter and initial business validation signed
M2	Requirements and scope baseline approved
M3	UX prototype and architecture approved
M4	Platform foundations operational
M6	Practicum-I gate: core marketplace prototype + KYC/payment adapters + AI PoC

Milestone	Deliverable / Gate
M7	Sitting and veterinary modules feature-complete
M8	Commerce and AI integration complete; code freeze for UAT
M9	UAT accepted and controlled Karachi pilot completed
M10	Pilot feedback/hardening complete
M11	Deployment and post-launch support gate
M12	Final closure, defense and archive

11.2 Schedule Management Rules

- Critical-path items are reviewed weekly.
- A task delayed >5 working days is escalated to the PM and reforecast.
- Schedule variance >10% at a major milestone triggers corrective action and supervisor visibility.
- Integration dependencies are started in parallel with development rather than at the end.
- Feature freeze applies before UAT; new scope is deferred unless formally approved.

12. RISK MANAGEMENT

Risk management covers project, technical, business, safety, privacy and external dependency risks. Probability and impact are rated 1-5. Risks scoring 16-25 are Critical, 10-15 High, 5-9 Medium and 1-4 Low. The register is reviewed bi-weekly and at every major gate.

ID	Risk	Domain	P	I	Score	Response	Mitigation	Owner
R1	Scope creep / over-complexity	Project	3	4	12	Mitigate	MoSCoW baseline; CR approval; defer advanced features	PM
R2	Insufficient sellers/sitters/vets/vendors for pilot	Business	4	5	20	Mitigate	Pre-launch supply onboarding and minimum pilot quotas	Business/PM
R3	Fraud / identity spoofing / misleading listings	Trust	3	5	15	Mitigate	Verification adapter, risk scoring, manual moderation, reporting	PM/QA
R4	Pet safety incident during sitting	Safety	2	5	10	Mitigate	Verified sitter workflow, emergency contacts, incident SLA, clear policy	PM
R5	Veterinarian non-adoption of health record workflow	Business	3	4	12	Mitigate	Co-design portal, minimal data entry, training and pilot with SME	PM/Vet
R6	AI gives irrelevant or unsafe guidance	AI/Technical	3	4	12	Mitigate	RAG + guardrails, red-flag rules, vet-reviewed benchmark, human escalation	AI/Vet
R7	Data breach / privacy exposure	Security	2	5	10	Mitigate	RBAC, encryption, secrets management, VAPT, minimum-data retention	QA/Security
R8	Payment/KYC API approval or downtime	External	4	4	16	Mitigate	Start onboarding early; adapter pattern; sandbox/mock fallback	Backend/PM
R9	Video consultation instability	Technical	3	3	9	Mitigate	Commercial SDK, reconnection, audio fallback, session reschedule	Dev
R10	Team resource bottleneck / academic workload	Resource	3	4	12	Mitigate	Cross-train, documentation, 15% schedule buffer, priority control	PM
R11	Regulatory/ethical concern over animal sales	Legal/Ethical	2	4	8	Mitigate	Terms, prohibited species policy, verification, vet-health disclosure	PM
R12	Vendor inventory / order inconsistency	Technical/Business	3	3	9	Mitigate	Inventory validation at checkout, vendor SLA, cancellation/refund flow	Dev/Business

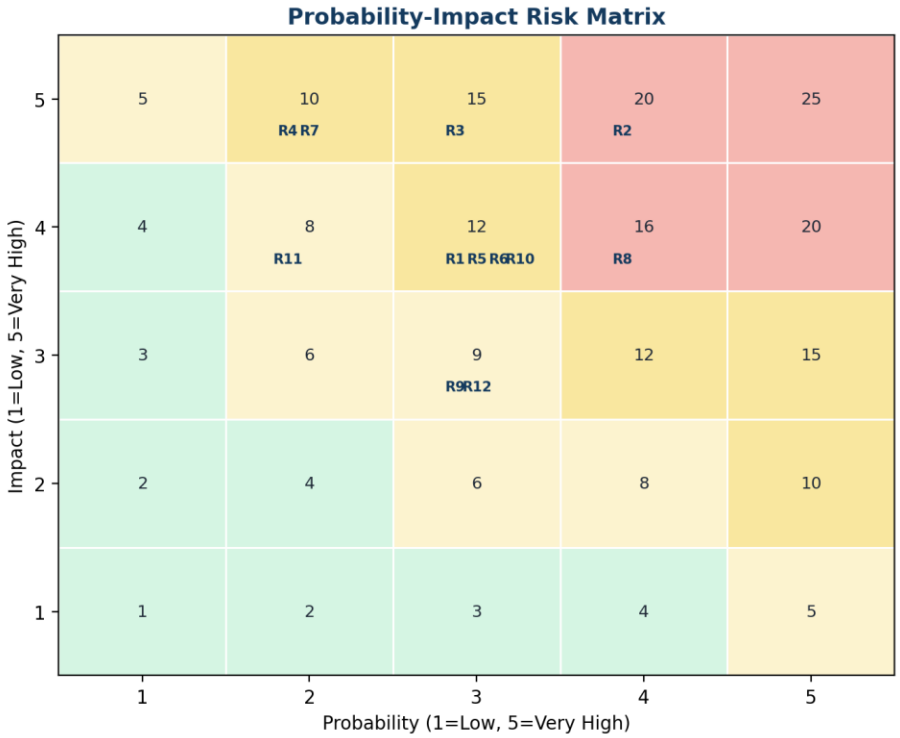


Figure 4. Probability-impact matrix showing the initial risk profile.

12.1 Risk Triggers & Contingency

Risk	Trigger	Contingency
R2 Supply liquidity	Pilot supply below minimum threshold two weeks before launch	Delay demand marketing; launch with one/two strongest verticals only.
R6 AI safety	Unsafe triage response found in benchmark/UAT	Disable generative answer; switch to deterministic routing + FAQ mode until fixed.
R8 External API	Provider access not available by M5	Use mock/sandbox adapter for practicum evidence; keep production release blocked.
R10 Team capacity	Critical-path owner unavailable >5 working days	Reassign tasks, reduce Should/Could scope, activate schedule buffer.
R7 Security	Critical/High VAPT issue	No-go for pilot until remediation and retest.

13. HUMAN RESOURCE MANAGEMENT

13.1 Resource Requirements

Resource	Count	Primary Responsibility
Project Manager / Team Lead	1	Planning, WBS/schedule, risk, stakeholder management, integration and academic reporting
Full-Stack / Backend Developers	2	Core services, APIs, database, payments/KYC adapters, web portals
Flutter / Frontend Developer	1	Mobile app user experience; may be combined with full-stack role in lean team
AI/ML Engineer	1	Recommendation engine, RAG assistant, risk scoring and model evaluation
UI/UX Designer	1	Research, flows, design system and usability testing; part-time
QA / Security Coordinator	1	Test planning, regression, security, UAT and VAPT coordination
DevOps / Cloud Engineer	1	CI/CD, environments, monitoring, backups and deployment; part-time
Veterinary SME	-1	Health data, triage guardrails, knowledge-base review and clinical UAT
Business / Partner Onboarding	1	Pilot supply recruitment, vendor/sitter/vet coordination and launch readiness

13.2 Lean Practicum Team Allocation

If the actual practicum team is three students, the formal roles above represent responsibilities rather than separate full-time people. The recommended allocation is: Student A - PM/Product/Business; Student B - Backend/DevOps/Integration; Student C - Frontend/AI/QA coordination, with cross-training and external SME support. Exact names should be added before submission.

13.3 Motivation Strategy

Herzberg Two-Factor: Maintain usable development environments, clear tools and requirements (hygiene), while giving ownership of meaningful modules and recognition for milestones (motivators).

McGregor Theory Y: Treat team members as self-directed professionals with autonomy over implementation choices within agreed architecture and acceptance criteria.

Goal-Setting Theory: Use specific measurable sprint and milestone goals, such as a working end-to-end marketplace flow or a defined AI benchmark target.

Expectancy Theory: Make the relationship between effort, completed evidence and practicum outcomes visible through transparent progress tracking and recognition.

14. TIME-PHASED BUDGET

14.1 Budget Assumptions

- Budget is an academic planning estimate, not a vendor quotation.
- Human resources are the largest cost driver; student labor may reduce actual cash outlay but is valued for project planning.
- Cloud/API/video/KYC/payment costs are estimated using sandbox/pilot usage.
- Security/VAPT and veterinary SME support are explicitly budgeted rather than assumed free.
- A contingency reserve is retained for rework, integration delays and paid tools.

14.2 Total Project Budget

Budget at Completion (BAC): PKR 4,800,000 (4.80 million / 48 lakh) academic planning baseline. This falls within the PKR 3.1M-5.5M investment range already identified in the Practicum Proposal.

Cost Category	Planned Cost (PKR)	Notes
Project management / product	500,000	Planning, governance, partner coordination
Software development	1,400,000	Backend/full-stack/mobile implementation
AI engineering	500,000	Recommendation, RAG assistant, risk scoring
UI/UX	250,000	Research, prototype and design system
QA & testing	300,000	Functional/UAT/test automation coordination
DevOps / cloud engineering	180,000	CI/CD, deployment and monitoring
Veterinary SME	180,000	Clinical requirements and AI safety validation
Business / onboarding	250,000	Pilot supplier/service-provider onboarding
Cloud, APIs, video, SMS	350,000	Pilot infrastructure and third-party usage
Security / VAPT	250,000	Independent/structured security assessment
Legal/compliance/partner setup	180,000	Policies, agreements, consultation
Pilot marketing / user acquisition	260,000	Controlled Karachi pilot
Contingency reserve	200,000	Integration, rework and unplanned tools
TOTAL	4,800,000	100%

14.3 Monthly Planned Value Baseline

Month	Primary Activities	PV (PKR)
M1	Initiation & validation	250,000

Month	Primary Activities	PV (PKR)
M2	Requirements & design	300,000
M3	Architecture & foundations	350,000
M4	Core platform development	450,000
M5	Marketplace/trust/AI build	550,000
M6	Practicum-I integration & AI PoC	650,000
M7	Sitting/vet/commercial modules	650,000
M8	AI integration, QA & code freeze	550,000
M9	UAT & Karachi pilot	400,000
M10	Hardening & documentation	250,000
M11	Deployment & support	200,000
M12	Closure & defense	200,000

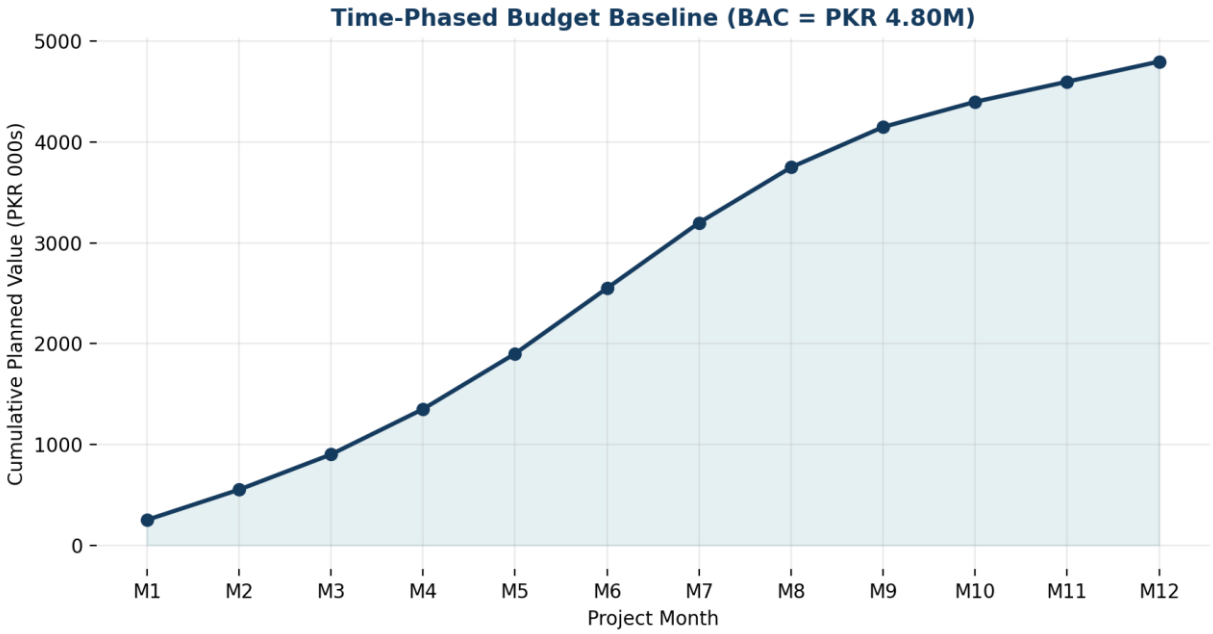


Figure 5. Cumulative planned-value baseline.

15. EARNED VALUE MANAGEMENT FRAMEWORK

Because this document is a pre-execution project plan, actual Earned Value (EV) and Actual Cost (AC) do not yet exist. Fabricating performance numbers would weaken the baseline. Instead, the project will use the approved monthly Planned Value (PV) distribution and populate EV/AC at monthly status dates during execution.

15.1 Baseline

BAC = PKR 4,800,000. Monthly PV values are defined in Section 14.3. A formal EVM review will be produced at the end of each month and at Practicum-I gate (end of Month 6).

15.2 EVM Terms & Formulas

Metric	Formula	Interpretation
PV	Budgeted cost of work scheduled	What should have been completed by status date
EV	Budgeted cost of work actually completed	Value of completed work
AC	Actual cost incurred	What has actually been spent
CV	EV - AC	Negative = over budget
SV	EV - PV	Negative = behind schedule
CPI	EV / AC	<1 = cost inefficiency
SPI	EV / PV	<1 = schedule inefficiency
EAC	BAC / CPI (if current cost trend continues)	Forecast final cost
ETC	EAC - AC	Forecast remaining cost
VAC	BAC - EAC	Expected final budget variance
TCPI	(BAC-EV)/(BAC-AC)	Required future cost efficiency to meet BAC

15.3 Practicum-I EVM Gate

At the end of Month 6, the team will calculate cumulative PV, EV and AC, then report CPI, SPI and forecast implications. Corrective action is mandatory if CPI or SPI falls below 0.90, or if the forecast exceeds the approved budget/schedule tolerance. Corrective options include reducing Should/Could scope, parallelizing QA/integration, limiting design rework, and switching AI modules to simpler rule-based fallback implementations.

16. COMMUNICATION MANAGEMENT

16.1 Communication Objectives

- Maintain transparent visibility of scope, schedule, cost, quality and risk while preserving auditable decisions and approvals.
- Escalate safety, privacy, security and AI issues immediately through the defined matrix.
- Use concise status reporting for academic stakeholders and visual prototypes/demos for non-technical partners.

16.2 Communication Matrix

Communication	Purpose	Audience	Frequency	Owner	Medium
Core team stand-up	Blockers and next 24-hour plan	Project team	Daily	PM/Lead	Teams/Meet/Discord
Weekly sprint/status review	Progress, schedule, risks, demos	Project team + QA	Weekly	PM	Meeting + Jira report
Supervisor review	Milestones, decisions, scope/risk	Supervisor + team	Bi-weekly or per practicum schedule	PM	Meeting + concise status deck
Veterinary SME review	Health records, triage rules, AI safety	Vet SME + AI/dev	At design and UAT gates	PM/AI	Workshop/demo
Partner onboarding sync	Readiness of sitters/vets/vendors	Business + partners	Bi-weekly during pilot prep	Business lead	Meet/WhatsApp Business/email
Risk & issue review	Top risks, triggers, owners, actions	Core team	Bi-weekly	PM	Risk register
Change request review	Impact of proposed scope change	Supervisor/PM/affected leads	As needed	PM	CR form + email approval
UAT / go-no-go	Acceptance results and pilot approval	Supervisor, SME, PM, QA	Major gate	PM	Formal sign-off

16.3 Escalation Matrix

Severity	Example	Escalation	Target
Level 1 - Operational	Normal defect, content issue, minor UI blocker	Developer/owner -> PM if unresolved	Same day
Level 2 - Project	Milestone delay, dependency failure, repeated QA failure	PM -> Supervisor/lead advisor	Within 24 hours
Level 3 - Critical	Security breach, pet-safety incident, unsafe AI output, legal/compliance issue	Immediate PM + Supervisor + relevant SME; stop affected function	Immediate

16.4 Progress Monitoring & Reporting

- Jira/Kanban for stories, defects and sprint flow.
- Master Gantt for milestone and dependency tracking.
- RAG status (Green/Amber/Red) for scope, schedule, cost, quality, risk and external dependencies.
- Weekly status report: completed, next, top risks/issues, decisions needed.
- Monthly EVM report after baseline execution begins.
- Formal MoM and decision log for supervisor/partner meetings.

APPENDIX A. AI INTEGRATION & GOVERNANCE

A.1 Why AI Belongs in ThePetsVault

AI is valuable when it reduces search friction, helps moderators identify risk and guides users toward the correct service. It is not included merely as a branding feature. The design follows a "decision-support, not autonomous authority" principle, especially for veterinary use.

A.2 AI Use Cases

Use Case	Value	MVP Approach
Personalized pet discovery	Rank listings using user preferences, location, species/breed interest, budget and verified-status signals.	Content-based ranking first; collaborative signals later.
Sitter matching	Rank sitters by date availability, distance, pet type/size experience, rating, price and owner preferences.	Rules + weighted scoring; learn weights after pilot data.
Product recommendation	Recommend food/accessories based on species, age, size, selected needs and purchase behavior.	Constraint-first rules + content similarity; exclude medical claims.
Grounded support chatbot	Answer platform/how-to questions and retrieve pet-care educational content from approved knowledge base.	RAG with citations internally and safe fallback.
Veterinary triage assistant	Collect symptoms, detect red flags and route to emergency/online/physical vet.	Deterministic red-flag rules + RAG; never diagnose.
Listing/fraud risk scoring	Flag suspicious prices, duplicate images, repeated account patterns and abnormal listing behavior.	Rules/anomaly score + human moderation.
Smart reminders	Vaccination/appointment/medication reminders based on structured records and vet-entered schedules.	Rules/automation rather than generative AI for reliability.

A.3 AI Safety & Governance Rules

- No AI-generated diagnosis, prescription or claim that a pet is medically safe.
- Red-flag symptoms trigger immediate escalation to a licensed veterinarian/emergency advice.
- The veterinary knowledge base must be reviewed and version-controlled by a veterinary SME.
- AI recommendations expose the main factors used for ranking where practical.
- Users can ignore/override recommendations; admins can override moderation flags.
- AI input/output logs used for evaluation must remove or minimize sensitive identity/payment data.
- Model/threshold changes are versioned and tested before production.
- If AI service is unavailable, the platform continues with normal search/filter and deterministic workflows.

A.4 AI Data Flow

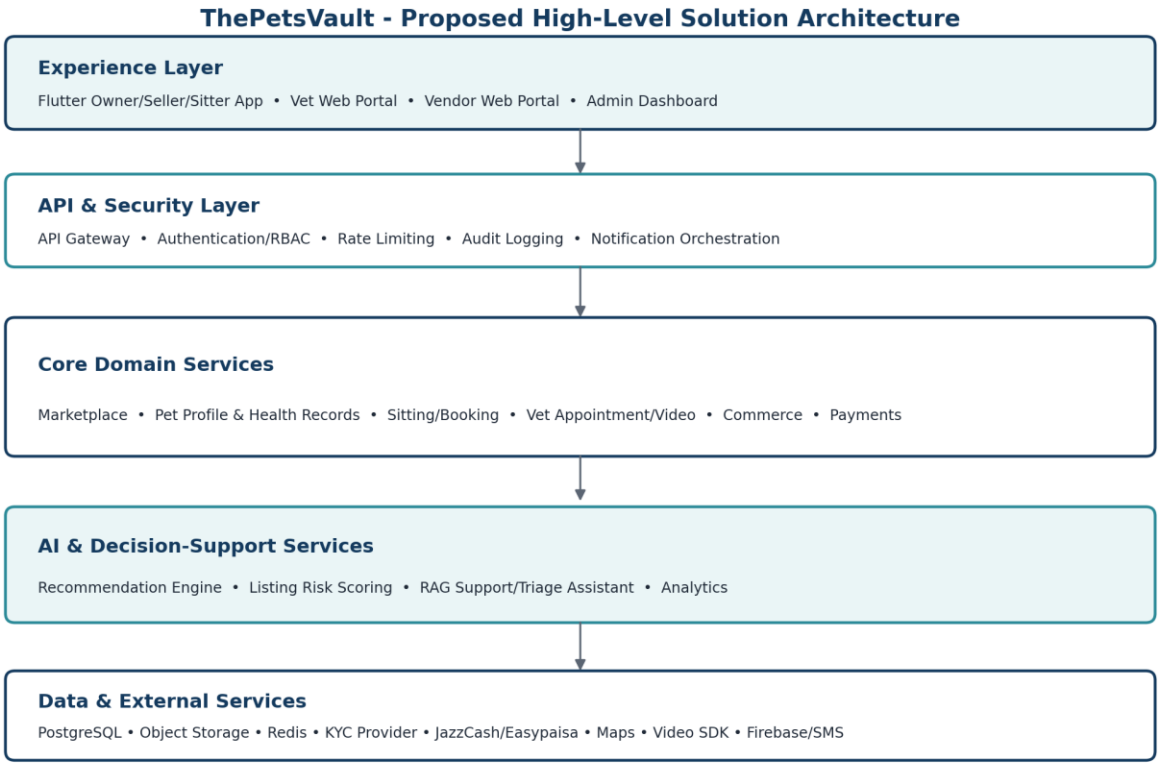


Figure 6. Proposed architecture with isolated AI and decision-support services.

A.5 AI Evaluation Plan

Stage	Dataset / Method	Evidence
PoC	Curated anonymized sample listings, sitter profiles, products and vet-reviewed triage scenarios	Notebook/API demo + metric report
Practicum-I Gate	Offline benchmark and supervisor/vet SME review	Precision/recall, response latency, unsafe-response test
UAT	Representative user tasks	User relevance scores, routing accuracy, false-positive review
Pilot	Opt-in interaction data and moderation outcomes	CTR/relevance, flags confirmed/overturned, escalation rates

APPENDIX B. MOSCOW PRODUCT BACKLOG

Priority	Backlog Item
Must	Authentication/RBAC and unified pet profile
Must	Pet buy/sell listings, search/filter, moderation and messaging
Must	Pet sitting profile, availability, booking and reviews
Must	Vet profiles, appointments, tele-consultation and health/vaccine records
Must	Food/accessory catalog, cart, checkout and vendor order workflow
Must	Payment/KYC provider adapters (sandbox/mock acceptable for practicum)
Must	Admin moderation, dispute/incident and analytics dashboard
Must	AI recommendation baseline + AI support/triage guardrails + listing risk score
Must	Security, logging, backup, test/UAT evidence
Should	Pet-specific product recommendations
Should	Sitter check-in photo/message updates
Should	Saved searches/favorites and recurring product reminders
Should	Automated vaccination/appointment reminders
Should	Basic promoted/featured listings
Could	Subscription/Plus membership
Could	Advanced collaborative recommendation model
Could	Pet microchip integration
Could	Adoption/NGO social-impact module
Could	Insurance partner integration
Won’t (MVP)	AI medical diagnosis / predictive health analytics
Won’t (MVP)	Owned warehouses, pet boarding facilities or delivery fleet
Won’t (MVP)	Custom payment rail / regulated escrow custody
Won’t (MVP)	Nationwide launch before Karachi pilot
Won’t (MVP)	Protected wildlife or illegal animal trading

APPENDIX C. DEFINITION OF DONE & QUALITY GATES

C.1 Definition of Done

- Acceptance criteria met and peer-reviewed code merged.
- Automated/unit tests passed where applicable.
- QA functional test passed on supported client/browser/device.
- Security/privacy checks completed for any feature handling identity, location, health or payment data.
- Payment/KYC/video integration tested in sandbox with failure/timeout cases.
- AI feature evaluated against defined benchmark and safety guardrails.
- No unresolved Critical/High defect for release candidate.
- API/user/admin documentation updated.
- Feature deployed to staging and demonstrated at sprint review.

C.2 Quality Gates

Gate	Entry Condition	Exit Evidence
Requirements Gate	Business validation complete	Signed scope/SRS and acceptance criteria
Design Gate	Requirements approved	Architecture, data model, UX prototype sign-off
Practicum-I Gate	Core prototype built	Marketplace demo, integration adapters, AI PoC metrics, updated plan
Code Freeze Gate	Core modules integrated	Regression pass, performance baseline, no critical defects
UAT Gate	Stable release candidate	UAT sign-off and resolved critical findings
Pilot Go/No-Go	Security and operational readiness	VAPT clearance, privacy/incident checklist, supervisor approval
Closure Gate	Pilot/support complete	Final report, demo, lessons learned, handover/archive

APPENDIX D. GLOSSARY & SOURCE MATERIALS

D.1 Glossary

Term	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BAC	Budget at Completion
BRD	Business Requirements Document
CPI	Cost Performance Index
EVM/EVA	Earned Value Management / Analysis
KYC	Know Your Customer / identity verification
MVP	Minimum Viable Product
PM	Project Manager
RAG	Red-Amber-Green project status
RAG (AI)	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
SIT	System Integration Testing
SPI	Schedule Performance Index
SPM	Software Project Management
UAT	User Acceptance Testing
VAPT	Vulnerability Assessment and Penetration Testing
WBS	Work Breakdown Structure

D.2 Source Materials Used to Prepare This Plan

- Practicum Proposal.pdf - project vision, four-pillar concept, target users, business model, technology options, budget range, milestones, risks and Agile/PMI perspective.
- Temp Hammad FAST Project.docx - ThePetsVault branding and draft SPM content, 12-month planning concept, WBS direction, stakeholder/HR/communication material and trust-layer ideas.
- SPM PROJECT_LRZK - AI-Powered Event & Wedding Management Marketplace.docx - reference structure for comprehensive SPM sections including feasibility, objectives, deliverables, scope, stakeholders, WBS, schedule, risk, human resources, time-phased budget, EVM and communication.

Academic note: Any external market-size, growth-rate, competitor or regulatory claims in the final practicum

report should be supported with formal published citations. This project plan intentionally focuses on the planning baseline and uses the supplied proposal as preliminary research input.

SUPERVISOR FINAL PLAN SIGN-OFF

Project Name: ThePetsVault - AI-Powered Pet Care & Marketplace Ecosystem

Decision on Practicum-I Project Plan Baseline:

Approval Status	☐ Approved ☐ Approved with Conditions ☐ Rework Required

Conditions / Comments:

_____

_____

_____

_____

Supervisor Name: _____

Signature: _____ Date: _____

Student / Team Lead Acknowledgement: _____ Date: _____